- **NUMPY**

```
import numpy as np
students=np.array([
    [85,90,78],    # student 1
    [60,55,70],    # student 2
    [95,92,88],    # student 3
    [32,30,34],    # student 4
    [72,68,75]
    ])
students
```

```
array([[85, 90, 78],
       [60, 55, 70],
       [95, 92, 88],
       [32, 30, 34],
       [72, 68, 75]])
```

```
avg_marks=np.mean(students, axis=1)
print("Average Marks:",avg_marks)
```

```
Average Marks: [84.33333333 61.66666667 91.66666667 32.         71.66666667]
```

```
avg_marks=np.mean(students, axis=0)
print("Average Marks:",avg_marks)
```

```
Average Marks: [68.8 67.  69. ]
```

```
topper=np.argmax(avg_marks)
print("Topper Student:",topper)
```

```
Topper Student: 2
```

```
failed_student=np.where(avg_marks<50)
print("Failed Student:",failed_student[0]+1)
```

```
Failed Student: []
```

```
# performance report
for i, avg in enumerate(avg_marks):
  status="pass" if avg >50 else"Fail"
  print(f"Student {i+1}: Avg ={avg:2f} | {status}")
```

```
Student 1: Avg =68.800000 | pass
Student 2: Avg =67.000000 | pass
Student 3: Avg =69.000000 | pass
```

- **PANDAS**

```
# create data frame and series
import pandas as pd
data={
    'Name':['Suruchi','Rivanshi','Spruha','Siri'],
    'Age':[10,23,16,20],
    'City':['New York','Paris','London','Tuscany']
     }
df=pd.DataFrame(data)
print(df)
```

```
       Name  Age      City
0   Suruchi   10  New York
1  Rivanshi   23     Paris
2    Spruha   16    London
3      Siri   20   Tuscany
```

```
s=pd.Series([10,20,30])
print(s)
```

```
0    10
1    20
2    30
dtype: int64
```

```
# basic data exploration.
# df.head() # head() return first top 5 row 5
# df.tail() # tail() return last top 5 row 5
df.shape
```

```
(4, 3)
```

```
df.columns
```

```
Index(['Name', 'Age', 'City'], dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Name    4 non-null      object
 1   Age     4 non-null      int64
 2   City    4 non-null      object
dtypes: int64(1), object(2)
memory usage: 228.0+ bytes
```

```
df.describe()
```

	Age
count	4.000000
mean	17.250000
std	5.619905
min	10.000000
25%	14.500000
50%	18.000000
75%	20.750000
max	23.000000

```
df['Age'].unique()
```

```
array([10, 23, 16, 20])
```

```
# Data Selection and Filtering
df[['Name','Age']]
```

	Name	Age
0	Suruchi	10
1	Rivanshi	23
2	Spruha	16
3	Siri	20

```
# Row Selection
df.iloc[0:,:-2] # [row:row,col:col]
```

	Name
0	Suruchi
1	Rivanshi
2	Spruha
3	Siri

```
# Row Selection
df.iloc[0:3,1:4] # [row:row,col:col]
```

	Age	City
0	10	New York
1	23	Paris
2	16	London

```
# Row Selection
df.iloc[0:2,1:2] # [row:row,col:col]
```

	Age
0	10
1	23

```
# Row selection by condition
df[df['Age']<=25]
```

	Name	Age	City
0	Suruchi	10	New York
1	Rivanshi	23	Paris
2	Spruha	16	London
3	Siri	20	Tuscany

```
# Row selection by condition
df[df['Age']<=15]
```

	Name	Age	City
0	Suruchi	10	New York

```
# Row selection by condition
df[df['Age']<=25]
```

	Name	Age	City
0	Suruchi	10	New York
1	Rivanshi	23	Paris
2	Spruha	16	London
3	Siri	20	Tuscany

```
# Row selection by condition
df[df['Age']<=25]
```

	Name	Age	City
0	Suruchi	10	New York
1	Rivanshi	23	Paris
2	Spruha	16	London
3	Siri	20	Tuscany

```
# Row selection by condition
df[df['Age']>10]
```

	Name	Age	City
1	Rivanshi	23	Paris
2	Spruha	16	London
3	Siri	20	Tuscany

```
# ADD Column
df['Salary']=[50000,40000,60000,15000]
df
```

	Name	Age	City	Salary
0	Suruchi	10	New York	50000
1	Rivanshi	23	Paris	40000
2	Spruha	16	London	60000
3	Siri	20	Tuscany	15000

```
# ADD Column
df['Company']=['Apple','Wipro','Accenture','Google']
df
```

	Name	Age	City	Salary	Company
0	Suruchi	10	New York	50000	Apple
1	Rivanshi	23	Paris	40000	Wipro
2	Spruha	16	London	60000	Accenture
3	Siri	20	Tuscany	15000	Google

```
# ADD Column
df['Gender']=["Female","Female","Female","Female"]
df
```

	Name	Age	City	Salary	Company	Gender
0	Suruchi	10	New York	50000	Apple	Female
1	Rivanshi	23	Paris	40000	Wipro	Female
2	Spruha	16	London	60000	Accenture	Female
3	Siri	20	Tuscany	15000	Google	Female

```
# Update a column
df['Age']=df['Age']+1
df
```

	Name	Age	City	Salary	Company	Gender
0	Suruchi	11	New York	50000	Apple	Female
1	Rivanshi	24	Paris	40000	Wipro	Female
2	Spruha	17	London	60000	Accenture	Female
3	Siri	21	Tuscany	15000	Google	Female

```
# Update a column
df['Salary']=df['Salary']+10000
df
```

	Name	Age	City	Salary	Company	Gender
0	Suruchi	11	New York	60000	Apple	Female
1	Rivanshi	24	Paris	50000	Wipro	Female
2	Spruha	17	London	70000	Accenture	Female
3	Siri	21	Tuscany	25000	Google	Female

```
# Delete column
df.drop('Salary',axis=1,inplace=True)
df
```

	Name	Age	City	Company	Gender
0	Suruchi	11	New York	Apple	Female
1	Rivanshi	24	Paris	Wipro	Female
2	Spruha	17	London	Accenture	Female
3	Siri	21	Tuscany	Google	Female

```
# ADD the rows
df.loc[3]=['Siri',21,'Tuscany','Google','Female']
df.loc[4]=['Anshika',33,"Amravati","Apple","Female"]
df=df.drop_duplicates()
df
```

	Name	Age	City	Company	Gender
0	Suruchi	11	New York	Apple	Female
1	Rivanshi	24	Paris	Wipro	Female
2	Spruha	17	London	Accenture	Female
3	Siri	21	Tuscany	Google	Female
4	Anshika	33	Amravati	Apple	Female

```
# Grouping and Aggregation
df.groupby('City').agg({'Age':['mean','max']})
```

	Age	
	mean	max
City		
Amravati	33.0	33
London	17.0	17
New York	11.0	11
Paris	24.0	24
Tuscany	21.0	21

```
# Sorting data
df.sort_values(by='Age',ascending=True)
```

	Name	Age	City	Company	Gender
0	Suruchi	11	New York	Apple	Female
2	Spruha	17	London	Accenture	Female
3	Siri	21	Tuscany	Google	Female
1	Rivanshi	24	Paris	Wipro	Female
4	Anshika	33	Amravati	Apple	Female

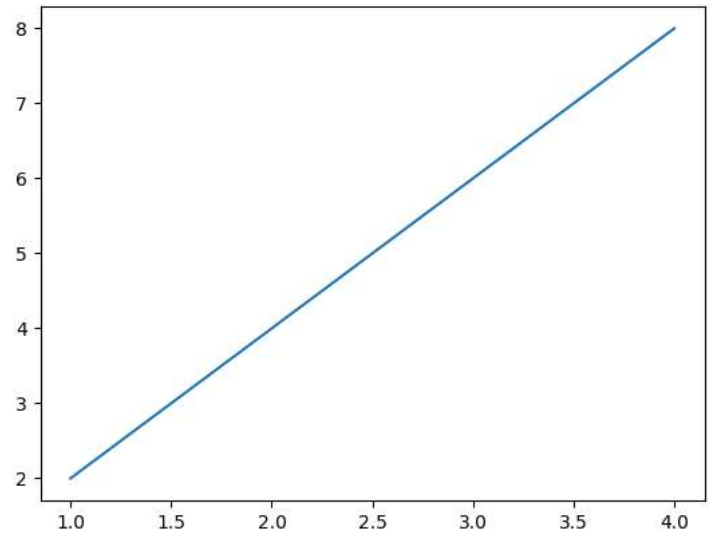
```
# value count
df['City'].value_counts()
pd.crosstab(df['City'],df['Age'])
```

Age	11	17	21	24	33
City					
Amravati	0	0	0	0	1
London	0	1	0	0	0
New York	1	0	0	0	0
Paris	0	0	0	1	0
Tuscany	0	0	1	0	0

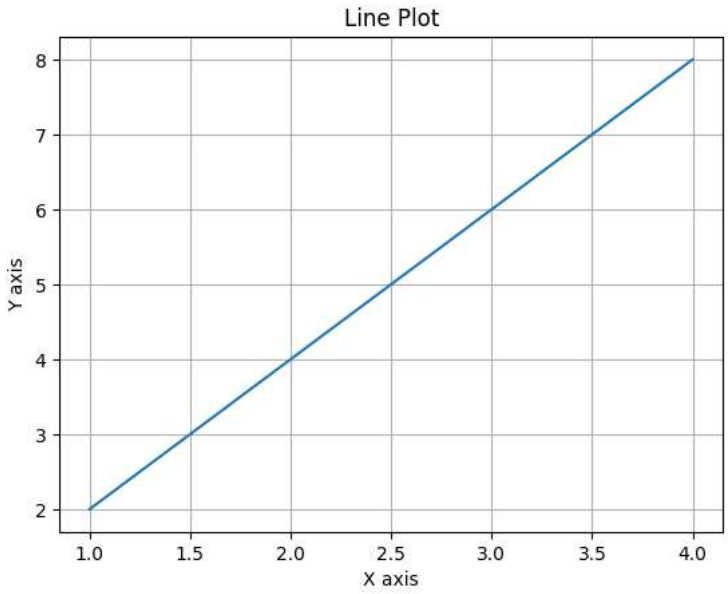
- **MATPLOTib**

```
import matplotlib.pyplot as plt
x=[1,2,3,4]
y=[2,4,6,8]
plt.plot(x,y)
```

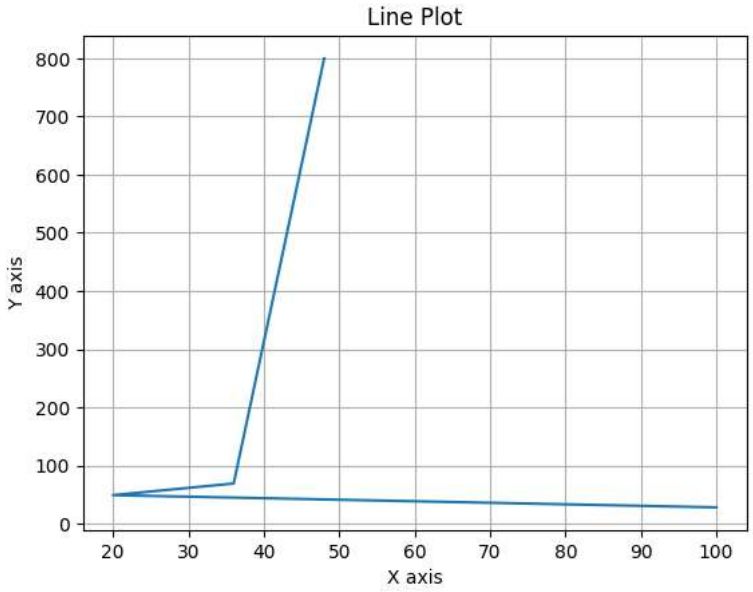
```
[<matplotlib.lines.Line2D at 0x7dfdf90b3d90>]
```



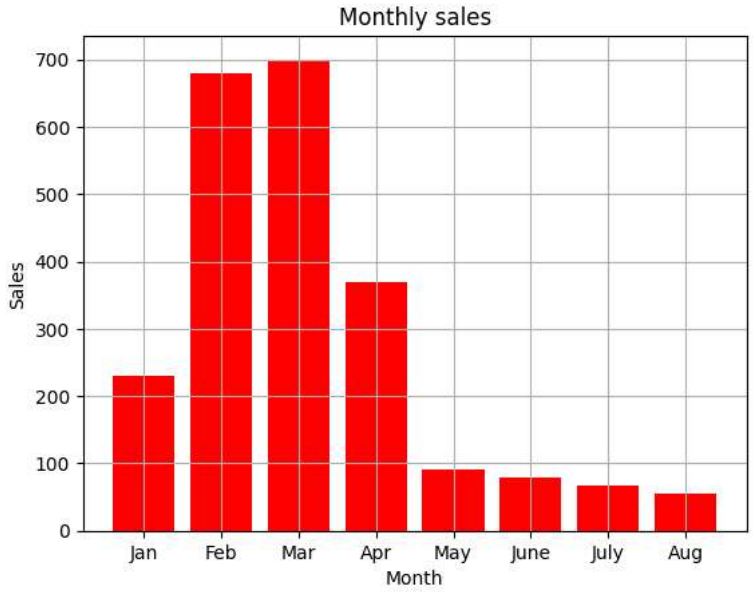

```
import matplotlib.pyplot as plt
x=[1,2,3,4]
y=[2,4,6,8]
plt.plot(x,y)
plt.title('Line Plot')
plt.xlabel('X axis')
plt.ylabel('Y axis')
plt.grid(True)
plt.show()
```



```
import matplotlib.pyplot as plt
x=[100,20,36,48]
y=[28,49,69,800]
plt.plot(x,y)
plt.title('Line Plot')
plt.xlabel('X axis')
plt.ylabel('Y axis')
plt.grid(True)
plt.show()
```




```
# Bar Chart
months = ['Jan','Feb','Mar','Apr','May','June','July','Aug']
sales=[230,679,700,370,90,78,67,56]
plt.bar(months,sales,color='Red')
plt.title('Monthly sales')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.grid(True)
plt.show()
```



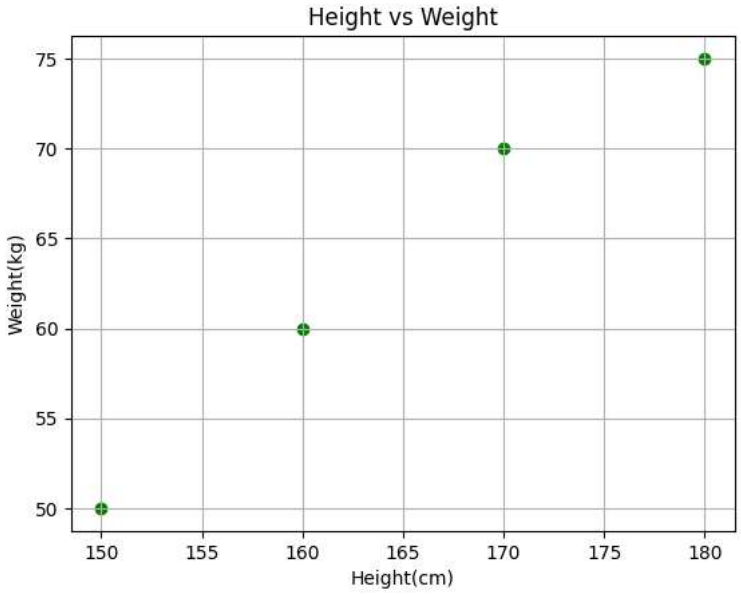

```
months = ['Jan','Feb','Mar','Apr','May','June','July','Aug']
sales=[230,679,700,370,90,78,67,56]

plt.barh(months,sales,color='Red')
plt.title('Monthly sales (Horizontal)')
plt.xlabel('sales')
```

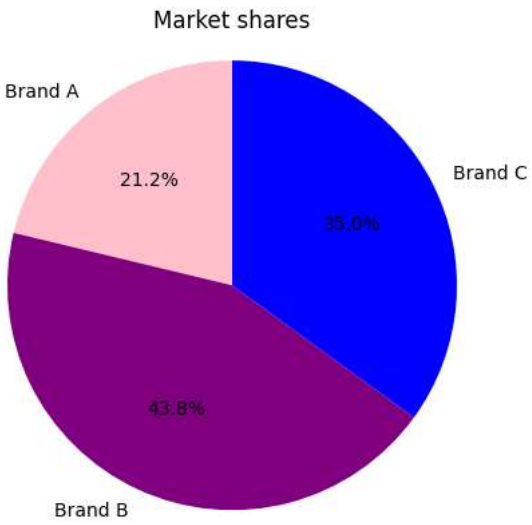
```
plt.ylabel('months')
plt.grid(True)
plt.show()
```



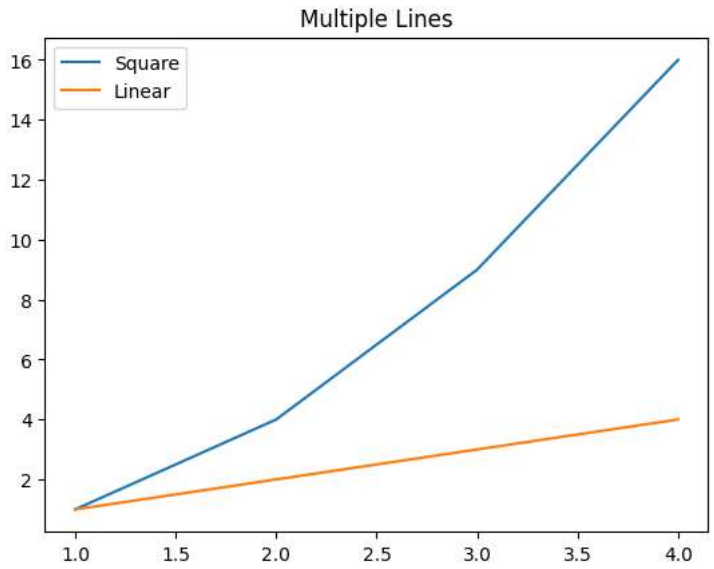

```
# Scatter Plot (comapring height vs weight)
height=[150,160,170,180]
weight=[50,60,70,75]
plt.scatter(height,weight,color='green')
plt.title('Height vs Weight')
plt.xlabel('Height(cm)')
plt.ylabel('Weight(kg)')
plt.grid(True)
plt.show()
```




```
# pie chart (market share)
brands = ['Brand A','Brand B','Brand C']
shares=[34,70,56]
plt.pie(shares, labels=brands, autopct='%1.1f%%',startangle=90,colors=['pink','purple','blue'])
plt.title('Market shares')
plt.axis('Equal')
plt.show()
```




```
# Multiple plot in one figure
x=[1,2,3,4]
y1=[1,4,9,16]
y2=[1,2,3,4]
plt.plot(x,y1,label='Square')
plt.plot(x,y2,label='Linear')
plt.legend()
plt.title('Multiple Lines')
plt.show()
```




```
# Subplots
fig, axs = plt.subplots(2,2)

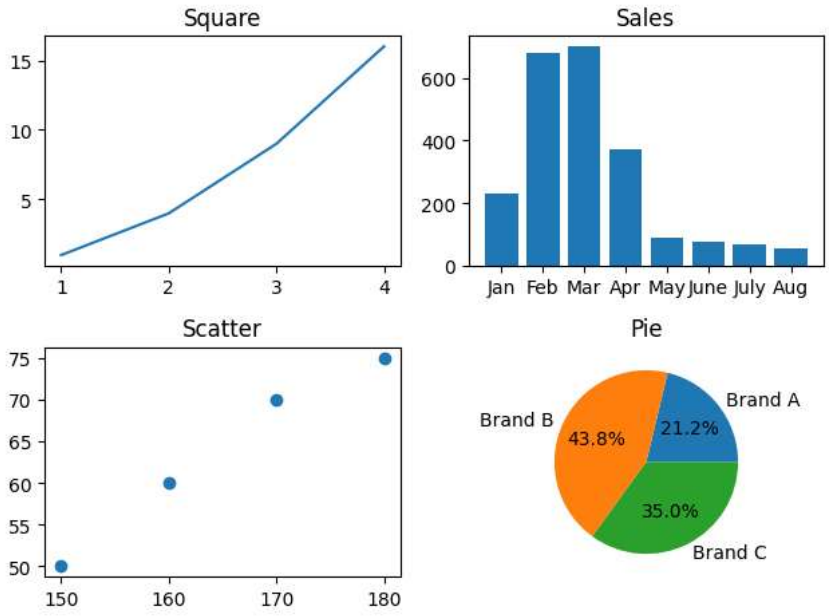
axs[0,0].plot(x,y1)
axs[0,0].set_title('Square')

axs[0,1].bar(months,sales)
axs[0,1].set_title('Sales')

axs[1,0].scatter(height,weight)
axs[1,0].set_title('Scatter')

axs[1,1].pie(shares, labels=brands, autopct='%1.1f%%')
axs[1,1].set_title('Pie')

plt.tight_layout()
plt.show()
```


Square
15
10
5
1
2
3
4
Sales
600
400
200
0
Jan
Feb
Mar
Apr
May
June
July
Aug
Scatter
75
70
65
60
55
50
150
160
170
180
Pie
Brand A
Brand B
43.8%
21.2%
35.0%
Brand C